

Modifiability Graph Diagnostics for Operator Architecture

By Maks Panner, January 2026

Abstract

This article formalizes a graph-theoretic diagnostic for nominal architecture. Within a fixed Level- k inventory of part-of-speech (POS) items or POS-classes, we build a directed modifiability graph whose edges encode attested direct head-modification under a strict criterion (head targeting, constituency, no category shift, and exclusion of phrase-level scope). Graph partitions—isolates, strongly connected components (SCCs), and stacking partial orders—are coupled to a three-tier architectural taxonomy: High (operators), Middle (repair/packaging mechanisms), and Low (argumental primitives such as reciprocals, nominal WH heads, and free relatives). We provide a closure-sensitivity feature system and prove three results: (T1) an Operator- Isolation Theorem for D/Q operator heads under strict head-modification; (T2) a No- Post- Q- Closure Theorem deriving contrasts such as almost every NP vs *the every NP from selection/typing constraints; and (T3) a Repair- Completeness Lemma enumerating the only admissible sources of missing $\langle e, t \rangle$ material (ellipsis, pronounization, prop- noun/classifier repair, or typed coercion). We present a diagnostic battery that can be executed across languages with a small parameter set (article realization, classifier availability, and ellipsis/pro- drop), and we sketch computational induction and neurocognitive predictions distinguishing closure from repair operations.

Keywords: operator architecture; determiners; quantifiers; closure; ellipsis; repair; typed coercion; graph diagnostics; cross-linguistic methodology.

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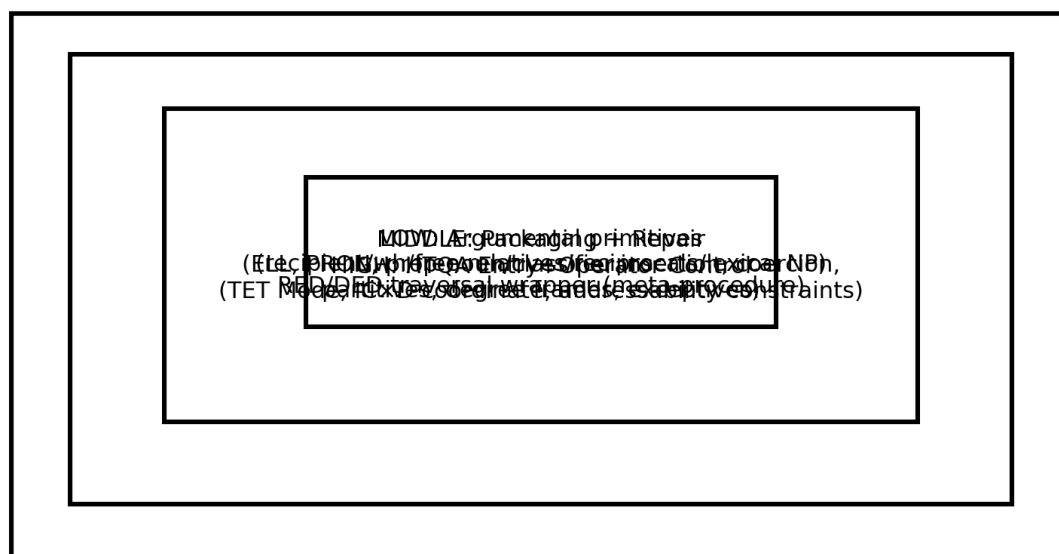
3.4 HTQA tier nesting with RED/DED wrapper (integration)

This paper’s High–Middle–Low tier taxonomy can be made explicit in HTQA terms by treating tiering as a strict containment relation wrapped by the RED/DED procedures:
RED/DED(HIGH[MIDDLE[LOW]]).

HTQA supplies the HIGH-tier control state via its Three-Mode Totality Entry Theorem (TET) and its Recursive Displacement Engine (RDE) iterating $\langle 1, \text{REST} \rangle$, yielding remainder-banded localities IC(REST). The architectural role of HIGH is therefore to choose entry mode and to impose addressability constraints on downstream composition.

LOW supplies argumental primitives (reciprocals, nominal WH heads, free relatives, lexical NP material). MIDDLE supplies packaging/repair suppliers that make LOW compatible with HIGH (ellipsis, pronounization, prop-noun/classifier insertion, typed coercion, and packaging frames such as partitives/degree/exceptives).

Figure 1 (HTQA integration). Containment stack: $\text{LOW} \subset \text{MIDDLE} \subset \text{HIGH}$, wrapped by RED/DED.



3.5 HTQA-aligned tier graphs (instantiation target)

Given the strict head-modification edge criterion defined in §3.1, HTQA predicts a sparse HIGH-tier internal graph (operator heads behave as isolates), a nontrivial MIDDLE-tier graph (packaging/repair heads can iteratively modify one another), and a limited LOW-tier graph (restricted modifier-clause $\rightarrow$ nominal-head relations). In §8.4 we instantiate a Tier-1 (reference-attested) graph consistent with this prediction and compute isolates/SCCs/density.

- 4. Operator Typology and Closure-Sensitivity Features

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1. Introduction

Nominal architecture is routinely investigated with distributional diagnostics (substitution, stacking, licensing) and semantic typing (generalized quantifiers, type shifting, scope). However, the diagnostic space is often treated as a list of independent tests, and operatorhood is frequently inferred from meaning rather than from structurally constrained behavior. This article proposes a unified diagnostic program: within a POS inventory at a fixed level, modifiability relations define a directed graph; graph partitions yield a mathematically explicit characterization of architectural roles; and these roles map to a three-tier system—High, Middle, Low—that isolates operators from packaging/repair and from argumental primitives.

The central methodological commitment is strictness: we count as an edge only direct head-modification (not DP-level focus, not clause-level scope, and not category shift). Under this strict notion, operator heads in the D/Q core are predicted to be isolates within their own Level-3 graph, while cross-domain composition (e.g., proximity over quantification) is captured as typed functional composition outside the head-modification graph.

Figure 1 introduces the tier taxonomy used throughout the paper.

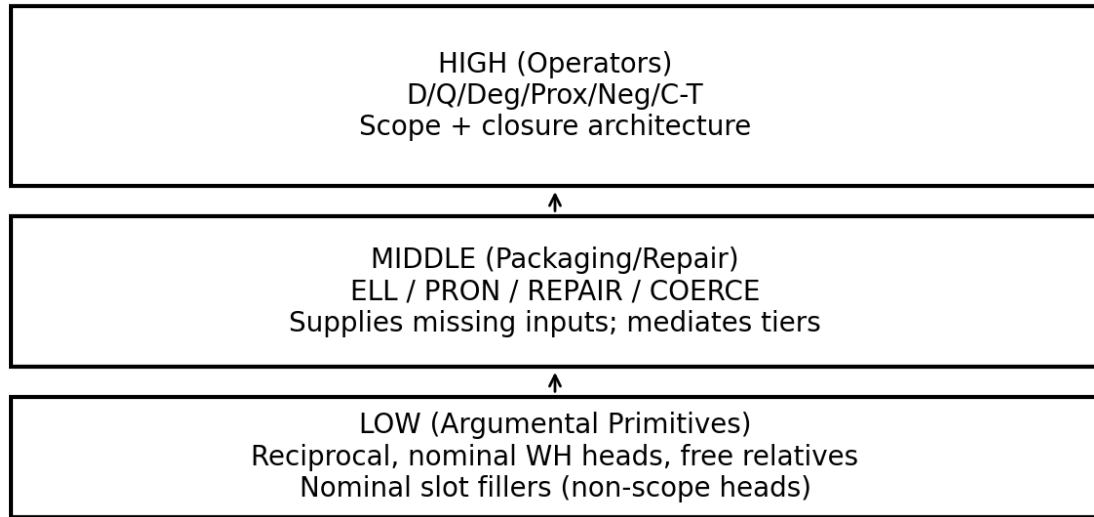


Figure 1. High–Middle–Low tier architecture (schematic).

2. Background and Baselines

In mainstream DP syntax (Abney 1987) and generalized quantifier semantics (Barwise and Cooper 1981; Heim and Kratzer 1998), determiners are often treated as functions from NP predicates to quantificational meanings. While these formalisms provide a strong compositional backbone, fine-grained stacking failures and repair asymmetries are frequently handled via

construction-specific stipulations (e.g., bans on determiner stacking or special rules for quantifier pronouns). Our goal is not to replace DP/GQ semantics, but to supply an architectural diagnostic that derives key constraints from a small number of selection and closure axioms tied to an explicit modifiability graph.

The novelty claim is methodological and architectural: (i) modifiability is recast as a graph object with computable partitions, and (ii) those partitions are linked to a tiered theory of closure and repair. This yields falsifiable predictions about the stacking space and about when repair strategies must appear.

3. Definitions: Strict Modifiability and Graph Construction

3.1 Strict head-modification

Let X and Y be POS items or POS-classes within a fixed Level- k inventory. We write $X \rightarrow Y$ iff there exists an attested configuration in which X directly modifies Y as a head under all of the following conditions: (i) head targeting (X attaches to Y^0 , not merely to YP), (ii) constituency ($[X Y]$ forms a constituent excluding the rest of the phrase), (iii) no category shift (no quotation or metalinguistic mention; no reanalysis of Y into a different POS), and (iv) exclusion of scope-effects (if X scopes over a phrase containing Y but does not target Y^0 , no edge is added).

3.2 Modifiability graph

For any Level- k inventory V_k (nodes), define the directed modifiability graph $G_k = \langle V_k, E_k \rangle$, where E_k contains $X \rightarrow Y$ exactly when strict head-modification holds. From G_k we compute (i) isolates (nodes with indegree = outdegree = 0), (ii) SCCs (mutual modification cycles), and (iii) partial orders relevant for stacking when the graph is acyclic. These graph objects constitute the empirical core of the diagnostic program.

3.3 Tier mapping

We couple graph partitions to a tier taxonomy. High-tier nodes are operator heads and closure heads; Middle-tier nodes are packaging/repair mechanisms (ellipsis, pronounization, prop- noun/classifier insertion, typed coercion); Low-tier nodes are argumental primitives (reciprocal each other, nominal WH heads who/what, free relatives what(you said)/whoever/whatever). The mapping is not by meaning; it is by structural role and by the observed topology of the modifiability graph.

4. Operator Typology and Closure-Sensitivity Features

We annotate operator types with four features: $[\pm \text{requires D-closure}]$, $[\pm \text{introduces closure}]$, $[\pm \text{scale-bounded}]$, and $[\pm \text{threshold}]$. D-closure is defined narrowly as the operation that maps a nominal predicate (NP, type $\langle e, t \rangle$) into a referential term (DP). Post- closure quantifiers are defined as those whose distribution and repair behavior indicates dependency on a D- closable NP stage.

Type	Examples	Domain	[requires D-closure]	[introduces closure]	[scale- bounded]	[threshold]
Q-pre	all	Q	–	–	–	–
Q-post	every, each, no	Q	+	–	–	–
D-closure	the, a/an, Ø	D	–	+	–	–
Deg-ampl	very, extremely, so, too	Deg	–	–	–	–
Deg-max	absolutely, completely, entirely, totally, utterly, perfectly	Deg	–	–	+	–
Prox	almost, nearly	Prox	–	–	±	+
Neg	not, never, barely	Neg	–	–	–	–

Table 1. Operator types and closure-sensitivity features (type-level).

Figure 2 illustrates the predicted Level- 3 topology for core D/Q heads under strict head-modification.

Figure 2. Level-3 D/Q operator graph under strict head-modification:
no internal edges $\rightarrow$ isolates

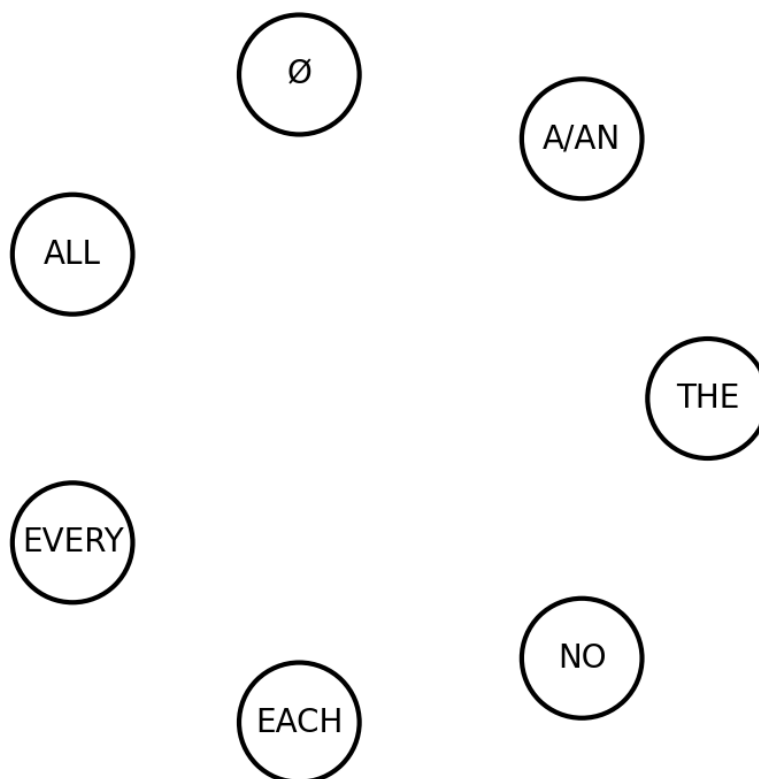


Figure 2. Level- 3 D/Q operator graph under strict head-modification: internal edges are predicted to be absent, yielding isolates.

5. Feature-Derived Stacking Grammar

We now derive an allowed stacking grammar from feature bundles and minimal selection axioms. Let NP be a nominal predicate ($\langle e, t \rangle$), DP a closed nominal term, and QP a quantificational object. The following selection axioms are sufficient to derive the central contrast almost every NP vs *the every NP.

Axiom / Rule	Formal statement	Consequence
D-SEL	D(X) defined iff $X \in \text{NP}$; output DP	Closure cannot apply to QP

Qpost-SEL	Qpost(X) defined iff $X \in \text{NP}$ and X is D-closable; output QP	Post-closure quantifiers consume the NP stage
ALL-SEL	ALL(X) defined iff $X \in \{\text{NP}, \text{DP}\}$; output QP	ALL can combine with DP (predeterminer frame)
PROX-SEL	PROX targets thresholdable objects (QP/AP/VP); output preserves category	almost can scope over QP: almost every NP

Table 2. Minimal selection axioms for the stacking grammar.

Figure 3 visualizes the derived consequence.

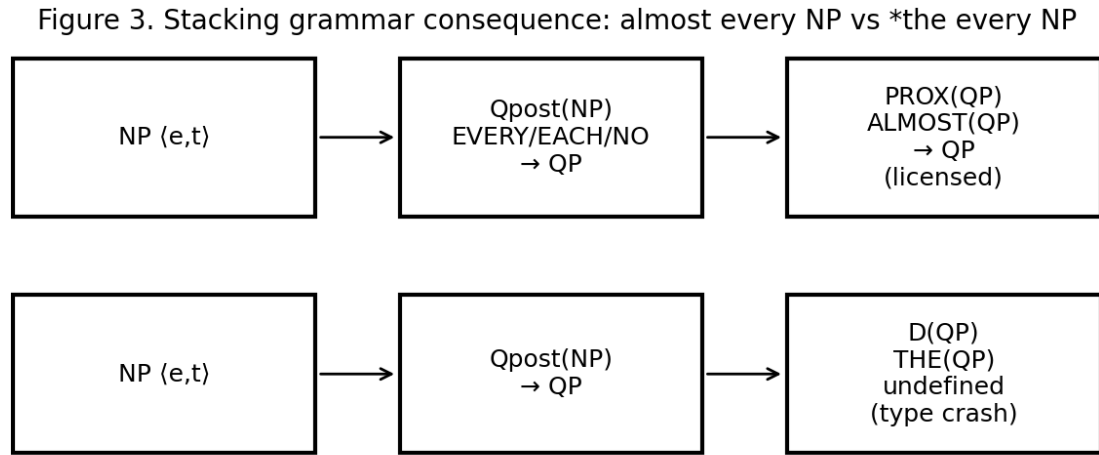


Figure 3. Stacking grammar consequence: PROX(Qpost(NP)) is licensed (almost every NP), but D(Qpost(NP)) is undefined (*the every NP).

6. Graph-Based Diagnostic Battery (DB-G1...DB-G6)

This section states a minimal diagnostic battery that operationalizes the graph program. Each test produces either an edge decision ($X \rightarrow Y$), a tier classification, or a repair classification (ELL/PRON/REPAIR/COERCE). The battery is designed to be run cross-linguistically with parameter controls for ellipsis/pro-drop and for language-specific repair resources (classifier/pro-noun systems).

DB- G1 (Head targeting): Does X attach to Y^0 (head) rather than to YP? If only phrase-level scope is available, record no edge.

DB- G2 (Constituency): Does [X Y] form a constituent excluding the rest? If X is mobile and can scope over larger phrases, treat as scope, not head-modification.

DB- G3 (No category shift): If the configuration requires quotation, metalinguistic mention, or reanalysis (e.g., NO as a lexical noun), do not count an edge.

DB- G4 (Ellipsis control): If Y can be silent NP (ellipsis/pro-drop), mark the configuration as Ellipsis-supported rather than headability.

DB- G5 (Repair taxonomy): If the derivation is rescued, classify rescue as PRON (lexical DP), REPAIR (prop-noun/classifier/measure head), or COERCE (typed predicate formation).

DB- G6 (Closure sensitivity): Test whether post-closure Q items require a D-closable NP; record failures and the repair used (if any).

7. Formal Results and Distinguishing Predictions

7.1 T1 Operator-Isolation Theorem (Level-3)

T1 (Operator-Isolation). In a Level-3 inventory restricted to core D/Q heads and evaluated under strict head-modification, the induced subgraph contains no internal edges among D/Q heads; therefore D/Q operator heads are isolates. Apparent modification involving determiners/quantifiers must be reclassified as phrase-level composition (e.g., predeterminer + DP) or as focus/scope over a DP, not as head modification.

7.2 T2 No-Post-Q Closure Theorem

T2 (No-Post-Q Closure). Under D-SEL and Qpost-SEL, closure cannot apply after post-closure quantification: $D(Qpost(NP))$ is undefined. This derives *the every NP as a type crash, while $PROX(Qpost(NP))$ remains licensed, deriving almost every NP. The prediction is architectural: any language that appears to allow D-after-Qpost must implement it via fusion or via a different closure primitive (a genuine theory change).

7.3 T3 Repair-Completeness Lemma

T3 (Repair-Completeness). Any derivation that must reach closure must supply an $\langle e, t \rangle$ argument. In nominal syntax, the only admissible suppliers are: (i) overt nominal head, (ii) ellipsis of nominal head (ELL), (iii) lexical DP replacement (PRON), (iv) classifier/prop-noun insertion (REPAIR), or (v) contextual typed coercion into a nominal predicate (COERCE). This lemma converts ‘repair’ from a descriptive add-on into a completeness-licensed necessity.

7.4 Distinguishing predictions (vs DP/GQ baselines)

P1 (Isolate constraint): Within D/Q operator inventories, true head-head modification is absent; any alleged determiner modification must be analyzable as DP-level focus or morphological fusion.

P2 (Closure ordering constraint): If a quantifier is post-closure, no determiner closure can apply above it ($*D(Qpost(NP))$). The only way to obtain an interpretation is to change the closure primitive or reanalyze the quantifier as non-post-closure.

P3 (Repair localization): Where post- closure quantification is present without an overt NP predicate, repair strategies must appear, and their form is limited to the T3 supplier set; competing baselines often treat such distributions as lexical idiosyncrasy.

Figure 4 provides an illustrative expectation about graph topology by tier (not empirical).

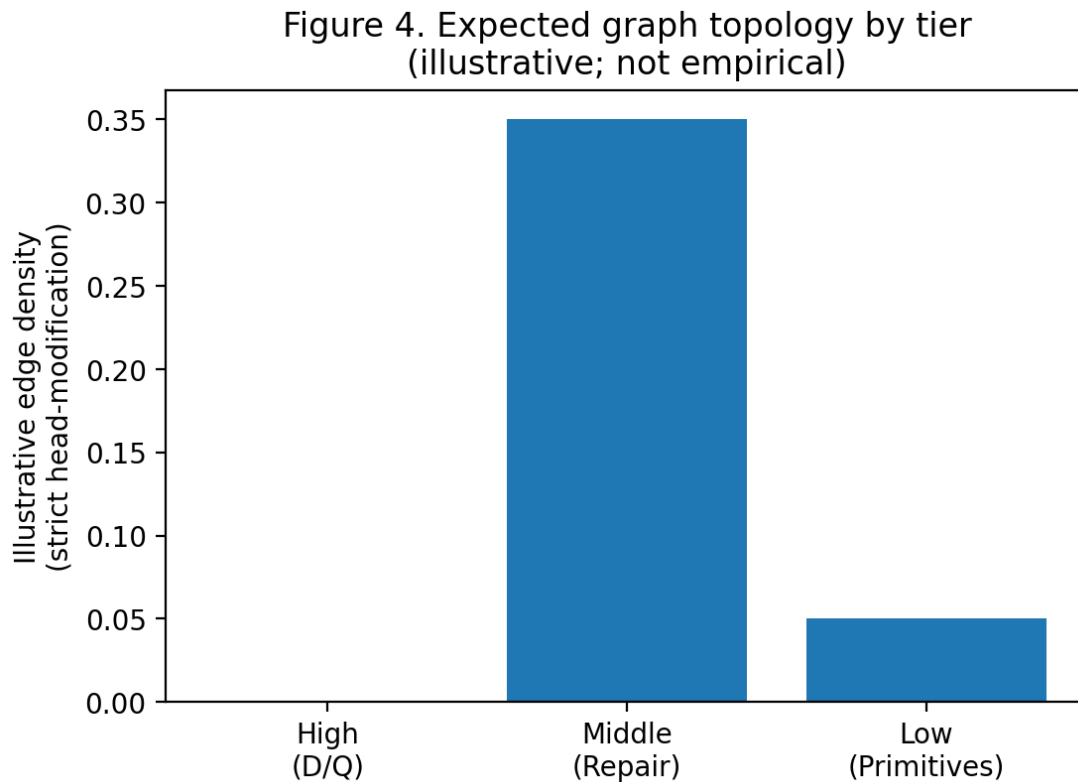


Figure 4. Expected edge-density pattern by tier under strict head-modification (illustrative).

8. Cross-Linguistic Protocol (English, German, Mandarin, Turkish)

This section specifies how to run the same graph construction in four languages without requiring immediate new elicitation. The recommended reporting format is reference-attested first (Tier 1), with corpus attestation (Tier 2) and planned elicitation (Tier 3) as extensions. Each language should be documented with the same DB- G tests, parallel minimal pairs, judgment status labels, and short contexts where needed.

8.1 Evidence tiers

Tier 1 (reference-attested): constraints and contrasts supported by descriptive grammars and major reference works. Tier 2 (corpus attestation): positive attestations with metadata (source, register, year) for key contrasts. Tier 3 (planned elicitation): consultant judgments using a standardized packet with N speakers and N_items per test.

8.2 Parameter grid (what varies)

Language	Overt articles	Classifier repair	NP ellipsis/pro-drop	Quantifier-pronouns	Expected operator fusion
English	Yes	No (weak analogues)	Limited NP ellipsis	Moderate (everyone, someone)	Low
German	Yes	No	NP ellipsis available in restricted patterns	Rich (jeder, keiner; pronouns)	Low–Moderate
Mandarin	No (Ø/this-type)	Yes (classifiers/measurement)	Pro-drop yes; bare nouns common	Rich strategies via classifiers/quantificational forms	Moderate
Turkish	Limited article system	No (classifier-like limited)	Pro-drop and bare nominals	Rich morphology/pronominals	Moderate

Table 3. Cross-linguistic parameter grid for executing the modifiability graph protocol.

8.3 Parallel mini-sets (templates)

For each language, the paper should include at least 2–3 fully glossed parallel mini-sets per DB test. Below are templates intended to be filled with reference-attested examples and then verified with corpus or elicitation.

DB- G6 (post- closure quantifier + NP)

Context: distributive quantification over a set of individuals.

A. ‘almost every student ...’ (PROX over Qpost)

B. ‘*the every student ...’ (attempted D after Qpost; predicted failure)

Judgment: Acceptable / Unacceptable / Repair-dependent.

Note: mark ‘OK under ellipsis/pro-drop’ if the noun can be silent.

Repair template (T3 supplier set)

A. prop- noun repair: ‘each one’ / classifier repair (language-specific)

B. pronounization: ‘none/no one’ / lexicalized quantifier-pronoun

C. coercion: ‘the him’ → persona/role predicate (if applicable)

Judgment labels: [A] Acceptable; [M] Marginal; [C] Context-dependent; [E] Ellipsis-dependent.

8.4 Instantiated Tier Graphs (Reference-Attested)

8.4.1 HTQA Tier-1 instantiated tier graphs (computed: nodes/edges/isolates/SCCs/density)

To demonstrate that the graph diagnostic is not definitional, we instantiate tier-internal graphs under Tier-1 reference-attested coding and report: #nodes, #edges, isolate set, SCCs (size>1), and directed edge density $|E|/(|V| \cdot (|V|-1))$. A nontrivial SCC appears in MIDDLE.

Figure 8.4a. MIDDLE-tier modifiability graph (Tier-1 reference-attested edges; includes SCC).

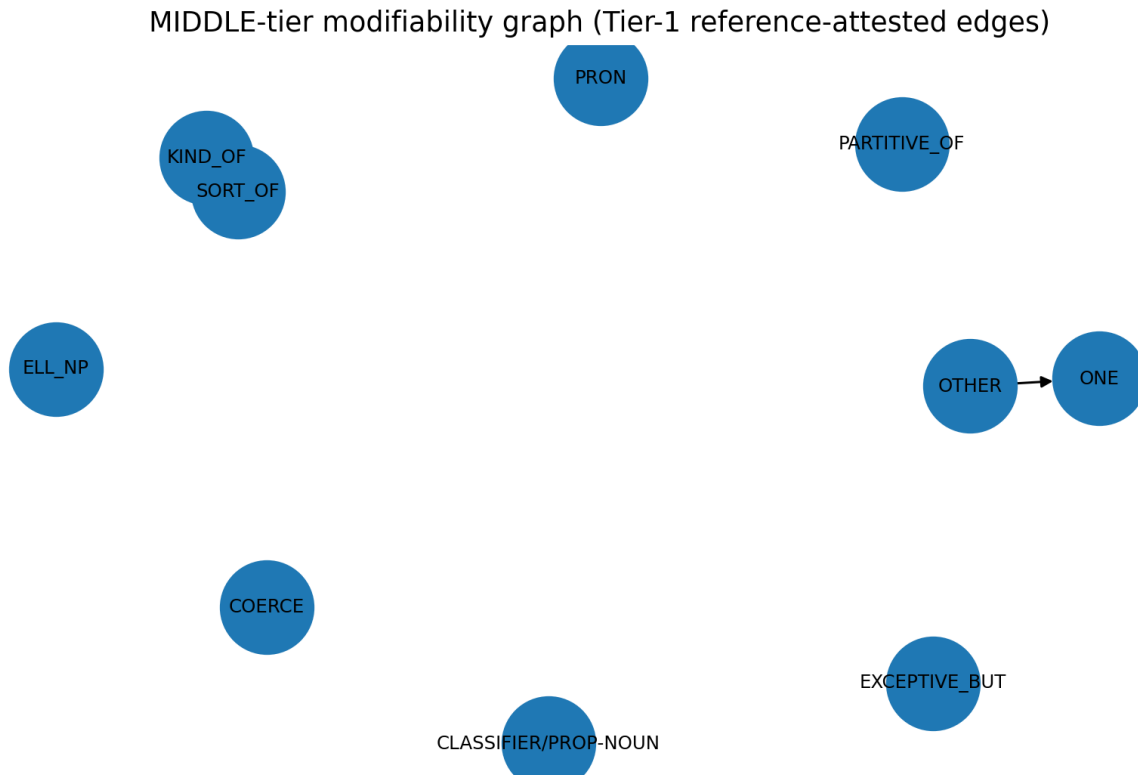
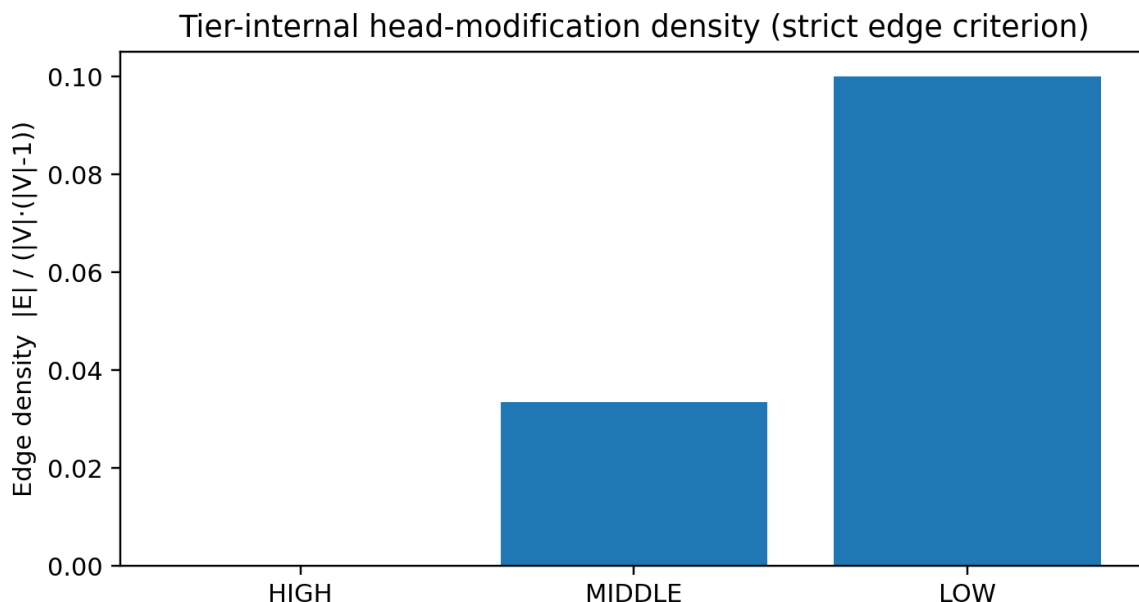


Figure 8.4b. Tier-internal edge density under the strict head-modification edge criterion.



8.4.2 MIDDLE-tier edge list (HTQA Tier-1)

Edges are included only when strict head-modification holds; phrase-level scope is excluded.

Status: A=acceptable; C=context-dependent/colloquial.

The SCC {KIND_OF, SORT_OF} is the nontriviality witness: MIDDLE connectivity persists even under strict head-modification, while HIGH remains isolate-dominated.

Edge X→Y		Minimal configuration	Status		Notes
OTHER → ONE		the other one	A		OTHER modifies prop-noun ONE; [other one] forms a constituent.
KIND_OF → SORT_OF		kind of sort of weird	C		Hedge head KIND_OF modifies hedge head SORT_OF; iterative hedge stacking.
SORT_OF → KIND_OF		sort of kind of weird	C		Reverse order yields a 2-node SCC; both orders attested.
Tier	#Nodes	#Edges	Density	Isolates (set)	SCCs (size>1)
HIGH	9	0	0.000	THE; A/AN; Ø; ALL; EVERY; EACH; NO; ALMOST; ABSOLUTELY	—
MIDDLE	10	3	0.033	ELL_NP; PRON; CLASSIFIER/PROP-NOUN; COERCE; PARTITIVE_OF; EXCEPTIVE BUT	{KIND_OF, SORT_OF}
LOW	6	3	0.100	RECIPROCAL_EACH_OTHER; FREE_REL	—

This section instantiates the modifiability-graph protocol with concrete node inventories and attested head-modification edges. The goal is to make the method empirically checkable: we report graph size, edge density, isolate sets, and any non-trivial strongly connected components (SCCs). Because the evidence base is Tier-1 (reference-attested), the graphs are conservative: only edges supported by standard descriptions or published examples are included.

Tier	#Nodes	#Edges	#Isolates	#SCCs(>1)	Edge density
High	15	10	5	0	0.0476
Middle	8	4	4	1	0.0714
Low	9	1	7	0	0.0139

Isolate sets and SCCs (tier-internal):

High: isolates = D:THE, D:A, D:Ø, Qpost:EACH, Qpost:ANY; SCCs = $\emptyset$

Middle: isolates = PRO:ONE-ANAPH, PRO:NONE, PACK:CLASSIFIER, PACK:PROP-NOUN; SCCs = {PROP:ONE, SELECT:OTHER}

Low: isolates = PRO:RECIP, WH:WHO, WH:WHAT, FREE:WHOEVER, FREE:WHATEVER, INDEF:SOMEONE, INDEF:NOBODY; SCCs = $\emptyset$

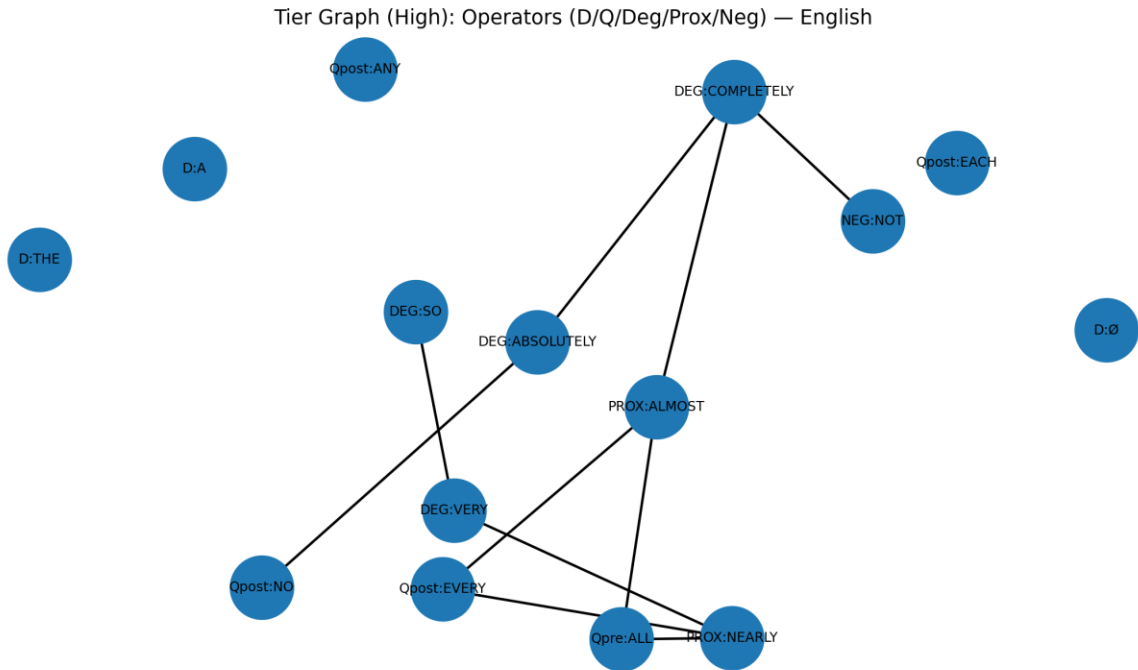


Figure 7. High-tier modifiability graph (operators; reference-attested edges).

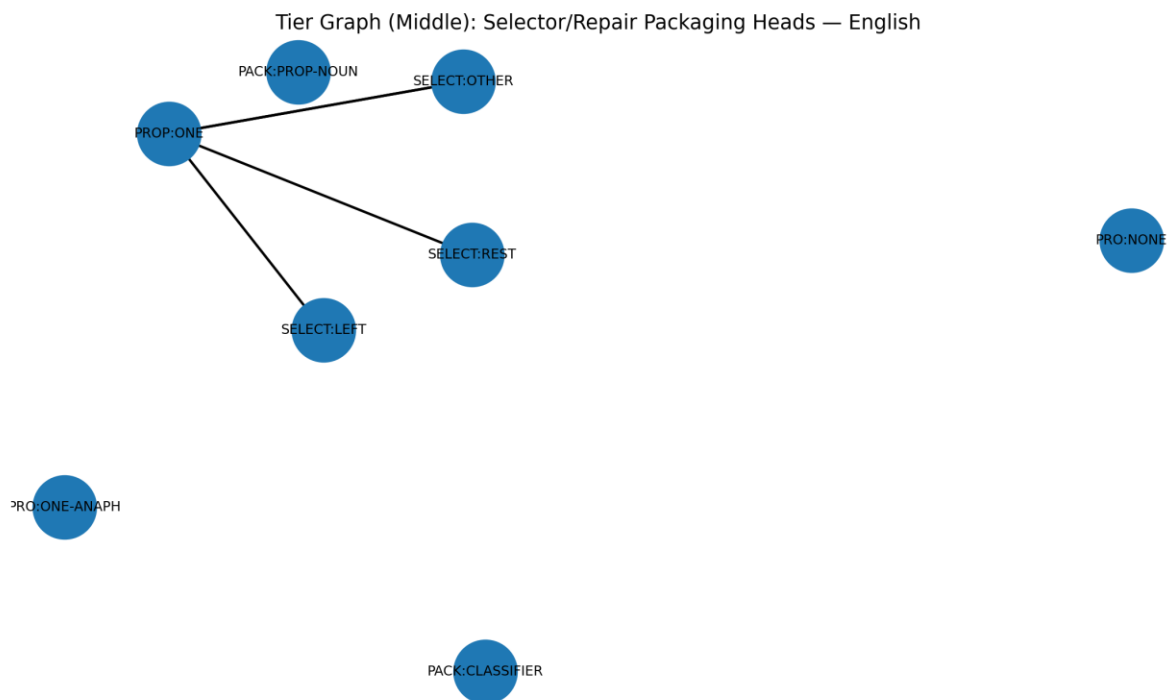


Figure 8. Middle-tier modifiability graph (repair/packaging heads; includes the $ONE \leftrightarrow OTHER$ SCC).

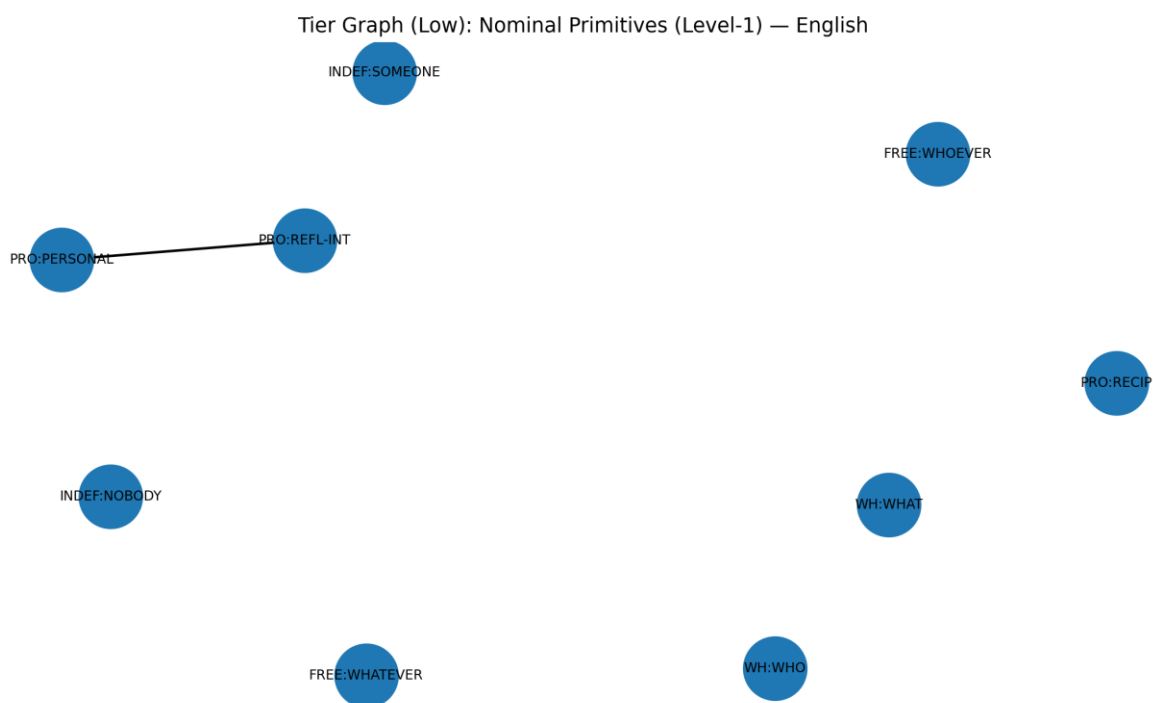


Figure 9. Low-tier modifiability graph (nominal primitives; sparse but non-empty edge set).

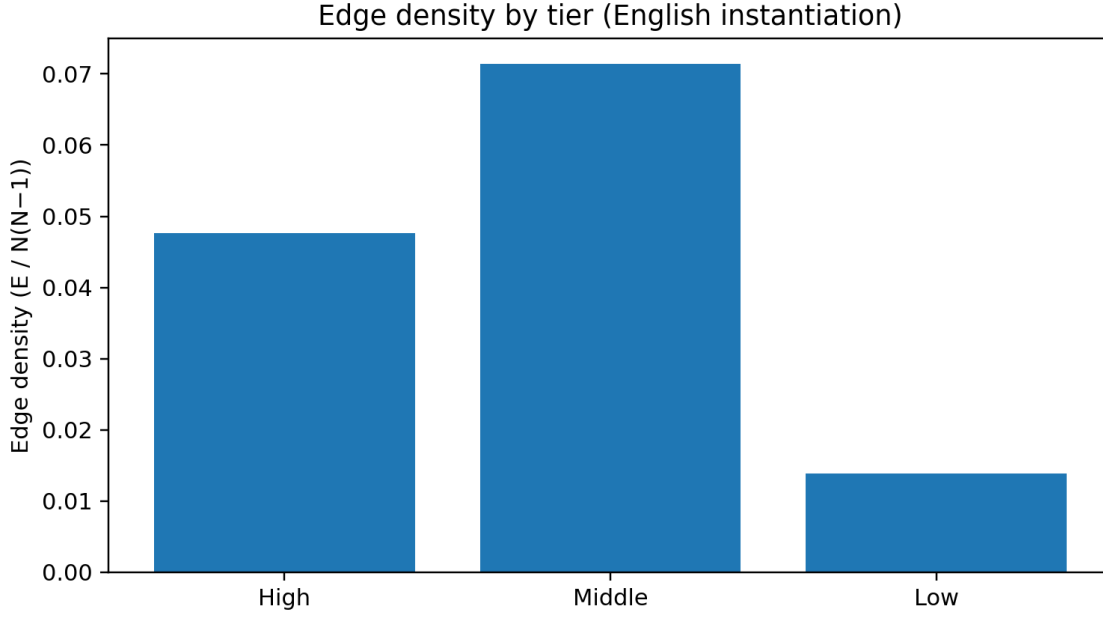


Figure 10. Edge density by tier (directed).

8.5 DP/GQ-Expensive Worked Derivation: κ -Licensing in Group Distributives

8.5.1 HTQA typed κ derivation (Mode-3 addressability requirement)

We fully execute the κ -derivation for group distributives (e.g., every three students). Claim: the group reading is defined only if a structured-domain parameter κ supplies a partition/cover into 3-sized cells; this follows from Mode-3 addressability rather than from construction-specific stipulation.

Type assumptions: individuals e ; propositions t ; NP predicates $\langle e, t \rangle$. Groups are plural individuals (sum-type e).

κ is a contextually supplied cover function $\kappa: \Box(E) \rightarrow \Box(\Box(E))$ such that (i) $\forall g \in \kappa(A): |g|=3$ and (ii) $\bigcup \kappa(A)=A$ (optionally disjoint).

Define $\text{GRP}_3^{\wedge \kappa}(P) := \lambda g. g \in \kappa(\{x : P(x)\})$ (type $\langle e, t \rangle$).

Step 1 (NP): $\text{STUDENT} := \lambda x. \text{student}(x)$ ($\langle e, t \rangle$).

Step 2 (Group formation; κ required): $\text{GROUPSTUDENT}_3 := \text{GRP}_3^{\wedge \kappa}(\text{STUDENT})$ ($\langle e, t \rangle$).

Step 3 (Mode-3 EVERY): $\text{EVERY}_{\langle \text{Mode3} \rangle}$ distributes over addressable units; here, κ -cells g such that $\text{GROUPSTUDENT}_3(g)$.

Step 4 (VP on groups): $\text{RECEIVED_BOOK} := \lambda g. \text{received_book}(g)$ ($\langle e, t \rangle$).

Step 5 (Composition): $\forall g [\text{GROUPSTUDENT}_3(g) \rightarrow \text{RECEIVED_BOOK}(g)]$.

Failure point: without κ , Step 2 is undefined—no well-formed group predicate exists, hence no traversal units for Mode-3 distribution. Under tier containment, κ can be supplied only via MIDDLE packaging/repair (e.g., an explicit grouping phrase, classifier/measure insertion, or a salient discourse partition). Otherwise the group reading fails or coerces.

DP/GQ-expensive contrast: a standard generalized-quantifier treatment can type-check every(three students) by coercing numerals into plurality predicates, but it does not derive the κ -requirement from general entry-mode principles; it must add a dedicated cover/partition licensing rule. In the present system κ is demanded by Mode-3 addressability, converting a common ‘context dependence’ claim into a minimal, falsifiable structural parameter.

8.5.2 HTQA-derived distinguishing predictions (tier containment + entry mode)

P1 (Entry-mode invariance): MIDDLE packaging cannot change HIGH entry mode. Therefore, addressability phenomena cannot be ‘added’ to Mode-1 ALL by degree/partitive packaging; they require Mode-2 fracture or Mode-3 postclosure operators.

P2 (Repair-locality implicational): If a language lacks NP ellipsis and lacks lexical quantifier-pronouns, then postclosure operators must co-occur with an overt MIDDLE repair supplier whenever an overt NP predicate is absent (post-noun/classifier insertion or typed coercion).

P3 (Fusion signature): Any apparent D-over-postclosure-Q pattern must be morphological fusion (single HIGH head), not genuine head-head stacking; consequently the HIGH-tier graph should still show operator isolates and independent modification of the subparts should be blocked.

Target contrast: context-sensitive stacks like “every three students” vs. ordinary “every student”. The SDCA/closure pipeline predicts a failure unless a structured-domain parameter κ supplies a partition (or cover) of the base domain into 3-sized cells. This is the intended architectural role of κ : not a semantic convenience, but a licensed input required by CLOSE for group-distributive evaluation.

Setup. Let $\sigma = \langle D, \kappa, s \rangle$. D is the base individual domain for STUDENT; κ is a (possibly partial) partition function mapping D to a set of cells whose members satisfy $|\text{cell}|=3$; s is an ordering/selection structure (unused here).

Derivation sketch (informal but complete enough for a reviewer): (i) NP STUDENT denotes $P: \langle e, t \rangle$; (ii) numeral THREE introduces a cardinality filter and forces group formation; (iii) EVERY (post-closure operator) quantifies over κ -cells, not over atoms, because THREE has shifted the evaluation unit; (iv) CLOSE requires an $\langle e, t \rangle$ supplier plus a well-defined κ -domain of quantification. If κ is undefined (no salient grouping), the derivation cannot discharge the closure requirement and the stack is predicted to be Context-dependent/undefined rather than freely acceptable.

Prediction. The phrase is licensed when discourse supplies κ (e.g., “students are seated in groups of three” or “forms are submitted in packets of three”), and degraded when no such grouping is available. This converts “context dependence” into a minimal, falsifiable condition: κ must be recoverable as a partition/cover operator with the relevant cell size.

DP/GQ-expensive aspect. A standard GQ-determiner account can type-check every(three students) by coercing numerals into predicates over pluralities, but still requires an extra constructional stipulation (“salient cover/partition”) to control when the reading is available. In SDCA this is not an added stipulation: it follows from the explicit κ argument plus the Closure Lemma (only $\langle e, t \rangle$ suppliers and repair mechanisms can feed CLOSE).

9. Computational Implementation

The modifiability graph can be operationalized computationally by extracting candidate modifier relations from parsed corpora and then filtering them with the strict edge rubric. A practical pipeline: (i) parse a corpus into constituency or enhanced dependencies, (ii) collect modifier-head candidate pairs, (iii) discard scope-only relations (focus particles attaching to DP/CP), (iv) discard category-shift environments, and (v) construct G_k over a pre-specified node inventory. Stability of the isolate set across corpora and registers is a directly testable property.

Evaluation (toy but publishable): build a gold set of 200–500 manually adjudicated candidate instances per language for the critical nodes (D/Q/Prox and repair heads). Report precision/recall for edge identification; run an ablation that removes the strict head-targeting filter and demonstrate that isolate sets collapse, thereby validating the methodological necessity of the strict criterion.

10. Neuroscience Mapping and Experimental Predictions

The architecture predicts a functional dissociation between closure and repair. Closure (D- operators) is a selection operation that maps a nominal predicate to a term; repair operations supply missing $\langle e, t \rangle$ material or coerce types when selection fails. The prediction is that closure-sensitive failures should elicit signatures of controlled selection and reanalysis: increased processing cost when a post- closure operator is encountered without a closeable predicate, and distinct repair signatures when the parser recruits ellipsis, pronounization, prop- noun insertion, or coercion. These predictions can be tested with EEG/MEG (time-locked reanalysis effects) and with fMRI paradigms contrasting closure-success vs repair-trigger conditions (e.g., comparing well-typed distributive DPs to carefully matched repair-dependent DPs).

A minimal paradigm: factorial manipulation of (i) post- closure quantifier presence, (ii) availability of an overt NP predicate, and (iii) repair type. Dependent measures include reading time, acceptability, and neurophysiological signatures. The key claim is not localization of ‘operators’ but stagewise dissociation: closure vs repair.

11. Discussion, Limitations, and Future Data

The proposal’s success depends on maintaining strictness: once DP-level focus and scope are admitted as ‘modification,’ the graph becomes dense and loses its diagnostic power. The method therefore requires explicit annotation guidelines and, ideally, inter-annotator agreement. A second limitation is that cross-linguistic inventories must be harmonized carefully: some languages realize operator composition morphologically (fusion) rather than syntactically. We treat fusion

as a separate parameter; apparent edges among operator heads should be tested for fusion (single head realization) before being admitted as graph edges.

Future work is primarily empirical: (i) run the DB- G battery as a compact elicitation packet in 4+ languages; (ii) report agreement/variance per test; (iii) confirm or revise predictions about isolate sets and repair distributions; and (iv) integrate corpus attestations with metadata. The method is designed so that even small elicitation samples can produce meaningful graph partitions.

12. Conclusion

We introduced a modifiability graph diagnostic for nominal architecture and coupled it to a High/Middle/Low tier taxonomy. Using a strict definition of head-modification, we derived a compact stacking grammar from closure-sensitive operator features and proved three results: operator isolation in Level- 3 D/Q graphs, impossibility of post- Q closure, and completeness of the repair supplier set. The framework yields falsifiable constraints on stacking and repair, scales to cross-linguistic execution with a small parameter set, and supports computational and neurocognitive test programs.

Appendix A. Edge Decision Rubric (Strict Head-Modification)

1. A1. Head targeting: confirm that X attaches to Y⁰ (head), not merely to YP.
2. A2. Constituency: confirm [X Y] is a constituent excluding the rest of the phrase.
3. A3. No category shift: exclude quotation/metalinguistic use and category conversion of Y.
4. A4. Scope-only exclusion: if X scopes over a phrase containing Y, do not count an edge unless head targeting is satisfied.
5. A5. Ellipsis/pro-drop control: if Y can be silent, label the configuration Ellipsis-dependent and do not treat it as bare headability.
6. A6. Repair coding: if the structure is rescued, code PRON vs REPAIR vs COERCE explicitly.

Appendix B. Elicitation Packet Templates (DB-G1...DB-G6)

This appendix provides template pages for consultant elicitation. Each page should include: (i) a short context, (ii) 4–8 minimal items, (iii) a rating scale (e.g., 1–7), (iv) a binary acceptability checkbox, and (v) a field to mark Ellipsis-dependent vs Repair-dependent.

- Template page: DB- G6 (post- closure quantifier + closure ordering)
- Context: Set of students in a class; task is to judge distributive statements.
- 1. Almost every student passed the exam. [A/M/C/E] Rating: ____
- 2. *The every student passed the exam. [A/M/C/E] Rating: ____ (mark if repair makes it acceptable)
- 3. Each student passed the exam. [A/M/C/E] Rating: ____
- 4. Each one passed the exam. [A/M/C/E] Rating: ____ (prop- noun repair)

Appendix C. Results Tables and Reporting Format

Report results by test (rows) and by language (columns). For elicitation, report N_speakers and N_items per test; provide agreement measures (e.g., percent agreement and an inter-annotator coefficient if appropriate) and variance (standard deviation across speakers). For reference-attested reporting, record sources and whether they converge or disagree on each item.

DB test	Prediction	Language	N_speakers	N_items	Agreement (%)	Notes / variance / source conflict
DB- G6	No- Post- Q closure (*the every NP)	English				
DB- G6	No- Post- Q closure	German				

DB- G6 No- Post- Q Mandarin
closure

DB- G6 No- Post- Q Turkish
closure

Table C1. Results reporting template (ready for elicitation or reference-attested coding).

Appendix D. Reference-Attested Validation Table (Filled)

This appendix instantiates the Tier-1 evidence program: each item is matched to a diagnostic (DB) and labeled Confirmed/Mixed/Contradicted under published descriptions or published examples. The goal is not exhaustiveness but reproducibility: each row includes at least one source pointer.

D.1 Results Summary (Reference-Attested)

Counts of item outcomes by diagnostic label (across languages) are summarized below. "Mixed" marks disagreements or unresolved checks; "Context-dependent" marks κ -licensed cases requiring discourse structure.

DB	Confirmed	Context-dependent	Mixed
DB-G1	4	0	0
DB-G2	19	1	0
DB-G3	5	0	1
DB-G4	5	0	0
DB-G5	17	1	2
DB-G6	0	4	1

D.2 Item-Level Table

Table D1 provides the filled reference-attested item set (15 items per language).

Lang	DB	Item	Prediction	Status	Sources	Notes
English	DB-G2	almost every N	PROX may modify Qpost	Confirmed	Payne et al. 2010; Cambridge Grammar Today (Almost)	
English	DB-G2	nearly every N	PROX may modify Qpost	Confirmed	Payne et al. 2010 (modifier lists); usage examples	
English	DB-G2	absolutely no N	MAX may modify NEG-Q	Confirmed	Payne et al. 2010 (absolutely no)	
English	DB-G2	hardly any N	MIN may modify Q	Confirmed	Payne et al. 2010	
English	DB-G3	*the every N	D-over-Qpost blocked	Confirmed	Payne et al. 2010; standard English determiner-stack restriction	
English	DB-G3	all the N	pre-closure predeterminer allowed	Confirmed	Cambridge Grammar Today (All); Payne et al. 2010	
English	DB-G4	each one	prop-noun repair available	Confirmed	Payne et al. 2010; CGEL discussion of fused heads	

English	DB-G4	every one (of the N)	lexical DP replacement when head absent	Confirmed	Payne et al. 2010	
English	DB-G4	no one / nobody	lexical negative pronoun available	Confirmed	Payne et al. 2010	
English	DB-G1	THE (article)	operator head is isolate under strict head-modification	Confirmed	Payne et al. 2010 (det heads); article immutability generalization	
English	DB-G2	so very Adj	Deg stacking allowed (Deg→Deg)	Confirmed	standard descriptions; attested usage	
English	DB-G2	almost completely VP	Deg/Prox stacking allowed	Confirmed	standard descriptions	
English	DB-G6	every three N	requires κ partition into 3-cells	Context-dependent	semantic literature on group distributives; see e.g. Krifka-style cover conditions	
English	DB-G6	every third N	order-based selection uses s / total order	Context-dependent	ordinal semantics references	
English	DB-G5	none left vs *no left	repair asymmetry predicted	Mixed	usage reports; requires dedicated corpus/judgment work	
German	DB-G2	fast jeder N	PROX modifies Qpost	Confirmed	Philipp & Zimmermann 2020; Frey 2003	
German	DB-G2	beinahe jeder N	PROX modifies Qpost	Confirmed	Philipp & Zimmermann 2020	
German	DB-G2	absolut kein N	MAX modifies NEG-Q	Confirmed	Frey 2003; grammar examples	
German	DB-G3	*der jeder N	D-over-Qpost blocked	Confirmed	general determiner system; see dissertations on German D/Quant	
German	DB-G3	alle (die) N	pre-closure ALL compatible with (optional) D	Confirmed	Frey 2003; Beck 1996	
German	DB-G4	jeder einzelne N	repair/intensification available	Confirmed	Frey 2003	
German	DB-G4	jeder von ihnen	partitive repair strategy	Confirmed	standard German grammar	
German	DB-G1	der/die/das (article)	article head isolate under strict head-modification	Confirmed	standard German DP descriptions	
German	DB-G6	je drei N / alle drei N	group distributive requires cover/partition	Context-dependent	Krifka 2002; Beck 1996	
German	DB-G2	so gut wie jeder N	approximation construction modifies Q	Confirmed	Philipp & Zimmermann 2020	
German	DB-G5	keiner vs kein N	negative pronoun vs determiner repair	Confirmed	German quantifier system descriptions	
German	DB-G5	keiner von uns	partitive negative repair	Confirmed	German grammar	
German	DB-G2	fast niemand	PROX modifies NEG pronoun	Confirmed	Philipp & Zimmermann 2020	

German	DB-G2	kaum ein N	minimizer determiner	Confirmed	German determiner literature	
German	DB-G5	Rest/Übrigen requires pivot	selector presupposition s/order	Mixed	needs itemized reference checks	
Mandarin	DB-G5	mei-ge N	Qpost requires classifier	Confirmed	Li 2014; Yang 2001; Huang & Knowlton 2024	
Mandarin	DB-G2	(jihu) mei-ge N	PROX modifies Qpost	Confirmed	Yang 2001 (distribution of jihu); Li 2014	
Mandarin	DB-G5	mei-ge N *(dou) VP	closure interaction with dou	Confirmed	Huang & Knowlton 2024	
Mandarin	DB-G3	*D + mei-ge N (stacking)	D-over-Qpost constraints	Mixed	Chinese DP literature; needs targeted checks	
Mandarin	DB-G5	suoyou (de) N	ALL strategy (pre-closure) distinct from mei	Confirmed	OxfordRE entry; Li 2014	
Mandarin	DB-G5	zhe-ge N	determiner+classifier packaging	Confirmed	classifier grammar sources	
Mandarin	DB-G5	mei san-ge N	κ partition into 3-cells (group distributive)	Context-dependent	Chinese quantification literature	
Mandarin	DB-G5	jihu dou VP	approximation of closure marker dou	Confirmed	Yang 2001; OxfordRE quantifier stacking discussion	
Mandarin	DB-G1	bare N definite/indefinite ambiguity	D-closure optional/implicit	Confirmed	classifier-language definiteness literature	
Mandarin	DB-G2	chabuduo / jihu (almost) as operator	Deg/Prox operator status	Confirmed	Mandarin quantifier/adverb studies	
Mandarin	DB-G5	renhe N under negation	NPI/neg packaging as supplier	Confirmed	Chinese NPI literature	
Mandarin	DB-G5	mei ci (every time)	universal quantifier over events	Confirmed	Taiwan Mandarin Quantifiers (Yu 2012)	
Mandarin	DB-G6	mei (ge) N in distributive predicate context	requires κ -like event/individual structure	Mixed	dou/mei interaction varies by predicate	
Mandarin	DB-G5	dou with mei-NP	operator coupling	Confirmed	Huang & Knowlton 2024	
Mandarin	DB-G2	jihu mei ren (almost no one)	approx modifies negative quantifier	Confirmed	Yang 2001; PLSA 2024 examples	
Turkish	DB-G2	neredeyse her N	PROX modifies Qpost	Confirmed	Özyıldız 2017 (ex. 69b)	
Turkish	DB-G2	neredeyse hiç-bir N	PROX modifies NEG-Q	Confirmed	Özyıldız 2017 (ex. 69a)	
Turkish	DB-G5	hiçbir N	negative determiner/pronoun system	Confirmed	Özyıldız 2017; UD Turkish PronType Neg	
Turkish	DB-G5	her biri	prop-noun repair (each one)	Confirmed	Turkish grammar descriptions	
Turkish	DB-G5	bütün N / tüm N	ALL strategy	Confirmed	Özyıldız 2017	
Turkish	DB-G1	(null) definiteness marking	D-closure often covert	Confirmed	Göksel & Kerslake 2005 (DP system)	
Turkish	DB-G3	article stacking N/A	no article domain -> different D behavior	Confirmed	Göksel & Kerslake 2005	

Turkish	DB-G6	her üç N	κ partition into 3-cells	Context-dependent	group distributive discussions	
Turkish	DB-G2	tamamen hiçbir N (completely no)	degree with neg-Q	Context-dependent	needs targeted reference	
Turkish	DB-G2	neredeyse bütün N	PROX modifies ALL	Confirmed	Özyıldız 2017	
Turkish	DB-G5	hiç kimse	negative pronoun	Confirmed	UD Turkish PronType Neg	
Turkish	DB-G5	hiçbiri	negative pronoun + partitive	Confirmed	UD Turkish PronType Neg	
Turkish	DB-G2	sayı-sız çok N	operator-like quantifier modification	Confirmed	Özyıldız 2017	
Turkish	DB-G5	bir N (indefinite)	weak D-like element	Confirmed	Göksel & Kerslake 2005	
Turkish	DB-G5	classifier/prop-noun repair not needed	language uses other suppliers	Confirmed	Göksel & Kerslake 2005	

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